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6) Alcohols

↳ Alcohols contain the hydroxyl unit as their functional group (OH)

↳ R – OH , R = Alkyl group

↳ Alcohols only have alkanes attatched to hydroxyl units.

↳ When naming Alcohols, replace the -e of the Alkane with -ol

↳ The lowest position number is given to the OH.

$CH_3 - OH$ = methanol

$CH_3 - CH_2 - OH$ = Ethanol

$CH_3 - \underset{\underset{OH}{|}}{CH} - CH_3$ = 2- propanol

Polynohols

↳ When there is more than 1 OH group.

↳ keep the -e of the alkane, and put a (di, tri, tetra) prefix before the -ol.

↳ Ex: 1,2 – ethandiol

↳ Ex: 1,2,3- Propane triol

Types of alcohols

1) Primary alcohol: alcohol in which the carbon that the OH is attatched to is only attatched to one other carbon.

↳ This is usually found 1st or last CH_3 or chain

↳ $\underset{\underset{OH}{|}}{CH_2} - CH_2 - CH_2 - CH_3$ 1- butanol

2) Secondary alcohol: In which the carbon with the OH group is attatched to only 2 Carbon atoms.

$CH_3 - \underset{\underset{OH}{|}}{CH} - CH_2 - CH_3$ 2 – butanol

3) Tertiary alcohol: alcohol in which the carbon attatched to the OH is attatched to 3 other carbons

$$CH_3 - \overset{\overset{CH_3}{|}}{\underset{\underset{OH}{|}}{C}} - CH_3$$

4) Cyclic alcohols: an alcohol that contains a cyclic ring.

5) Aromatic alcohols: An alcohol that contains a benzene ring. This is called a phenol. Used in plastic, drugs, dyes and weed killers.

Properties of alcohols

↳ The hydroxyl group is polar and allows for hydrogen bonding.

↳ Higher melting and boiling points of alcohol than the corresponding hydrocarbons due to intermolecular forces.

↳ These increase with chain length.

↳ alcohols are quite soluble in water

↳ As chain increases, solubility decreases.

C1 – C5 = highly soluble

C5 – C7 = moderately soluble

C8 → = slightly soluble / insoluble.

Reactions of Alcohol

↳ Preparing Alcohols: Hydration Reaction [Alkene → Alcohol]

$$\underbrace{\text{water molecule} + \text{alkene} \xrightarrow{\text{sulphuric acid}} \text{Alcohol}}$$

↳ Addition Reaction

Water molecule + alkene $\xrightarrow{\text{sulfuric acid}}$ Alcohol

↳ Addition Reaction
↳ Hydration Reaction because of the overall addition of water.
↳ Follows Markovnikov's rule: put hydrogen on the carbon atom with more hydrogen atoms.
↳ "Rich get Richer"

↳ Ex: $CH_3 - CH_2 - CH = CH_2 + HOH \xrightarrow{\text{acid}} CH_3 - CH_2 - CH(OH) - CH_3$

1-butene → 2-butanol

↳ Combustion of Alcohols: Alcohols combust to release $H_2O_{(l)}$, $CO_{2(g)}$ and energy

↳ Ex: $2\ C_3H_7OH_{(l)} + 9\ O_2 \longrightarrow 6\ CO_2 + 8\ H_2O$ + Energy

↳ Elimination Reactions: Alcohol → Alkene

↳ Alcohol can be decomposed to produce alkenes and water, in presence of sulfuric acid.
↳ Dehydration reaction.
↳ Alcohol $\xrightarrow{\text{Sulfuric acid}}$ Alkene + HOH

↳ Ex: $CH_3 - CH(OH) - CH_2(H) \xrightarrow{H_2SO_4} CH_3 - CH = CH_2 + H_2O$

2-Propanol → 1-Propene

↳ Substitution Reaction: Alcohol + hydrogen halide ⟶ alkyl halide + ter

↳ Ex: $CH_3 - CH_2(OH) + HCl \xrightarrow[\text{Lucas reagent}]{ZnCl_2} CH_3 - CH_2(Cl) + H_2O$

↳ $ZnCl_2$ (Lucas Reagent): A qualitive test to find the type of a alcohol
↳ happens because of difference in solubility
↳ Tertiary alcohol → Cloudy immediately
↳ Secondary alcohol → cloudy after 5 min
↳ Primary alcohol → Cloudy long after 5 min

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2) Ethers:
↳ Both parts of the Oxygen are bonded to the alkyl group, instead of just one, like in alcohols.

↳ R − O − R'

Properties of Ethers:

↳ much less polar than alcohols as they can't form hydrogen bonds
↳ good solvents as they mix both polar and non-polar substances.
↳ lower melting point and boiling point than alcohol but higher than corresponding hydro carbons.
↳ Chemically inert
↳ flammable

Naming

↳ 1) find the Oxygen
2) find the longest chain connecting to it.
3) Use that chain as the base name
4) the shorter chain becomes a branch and gains an OXY ending.

3) Use that chain as the base name
4) the shorter chain becomes a branch and gains an OXY ending.
5) make the oxygen link / ether bond as low as possible on the parent chain

Ex: $\overset{2}{CH_3} - \overset{1}{CH_2} - O - \overset{1}{CH_2} - \overset{2}{CH_2} - \overset{3}{CH_3}$

branch parent

1- ethoxypropane

Reactions:

↳ Condensation: Ethers from Alcohol.

↳ Ethers are formed when 2 alchols, in the presence of an acid react, and a water molecule is eliminated.

↳ This is called a condensation reaction.

$$\underset{\text{alcohol}}{R-OH} + \underset{\text{alcohol}}{HO-R'} \xrightarrow{H_2SO_4} \underset{\text{Ether}}{R-O-R'} + \underset{\text{water}}{H_2O}$$

$$\text{Ex: } \underset{\text{Methanol}}{CH_3-OH} + \underset{\text{Methanol}}{HO-CH_3} \xrightarrow{H_2SO_4} \underset{\text{Methoxy methane}}{CH_3-O-CH_3} + H_2O$$

Halides:

↳ A Carbon and Hydrogen compound in which 1 or more hydrogen atoms have been replaced by halogen atoms.

↳ Naming organic halides -

1) The halogen atom is considered as an attatchment on the parent chain.

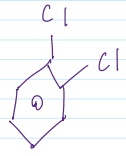
2) The name is shortened to fluro-, chloro-, bromo-, iodo-.

i: R- X

X = Cl, F, Br, I

3) Double bond takes priority over halogen

Ex:



1, 2- dichloro benzene

2, 2, 5 - tri bromo - 5- methyl hexane

$CH_3 - C - CH_2$

Properties:

↳ More polar

↳ higher intermolecular forces

↳ Organic halides have higher boiling points than corresponding hydrocarbons